

Foundations of Hierarchical clustering

[Hierarchical clustering]

1.0 Content Block

Complexity The standard algorithm for hierarchical agglomerative clustering (HAC) has a time complexity of $O(n^3)$ and requires $O(n^2)$ memory, which makes it too slow for even medium data sets. However, for some special cases, optimal efficient agglomerative methods (of complexity $O(n^2)$) are known: SLINK for single-linkage and CLINK for complete-linkage clustering. With a heap, the runtime of the general case can be reduced to $O(n^2 \log n)$ instead of $O(n^3)$, at the cost of additional the memory requirements. In many cases, the memory overheads of this approach are too large to make it practically usable. Methods exist which use quadrees that demonstrate $O(n^2)$ total running time with $O(n)$ space. Divisive clustering with an exhaustive search is $O(2^n)$, but it is common to use faster heuristics to choose splits, such as k-means.

2.0 Content Block

In this visualization, each point begins as its own cluster. At each step, the two clusters with the smallest distance between them are merged. On the left, dashed lines indicate which clusters are combined, and a running list of cluster groupings shows how clusters evolve over time. On the right, the dendrogram is constructed simultaneously, with each merge represented by a new branch. The height of each merge in the dendrogram corresponds to the distance or dissimilarity between the clusters being combined. Merges that occur at lower heights represent clusters that are more similar, while merges at higher heights indicate greater dissimilarity. Color-coding is used to associate each merge step in the data visualization with its corresponding branch in the dendrogram.

3.0 Content Block

Some of these can only be recomputed recursively (WPGMA, WPGMC), for many a recursive computation with Lance-Williams-equations is more efficient, while for other (Hausdorff, Medoid) the distances have to be computed with the slower full formula. Other linkage criteria include:

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ALGLIB implements several hierarchical clustering algorithms (single-link, complete-link, Ward) in C++ and C# with $O(n^2)$ memory and $O(n^3)$ run time. ELKI includes multiple hierarchical clustering algorithms, various linkage strategies and also includes the efficient SLINK, CLINK and Anderberg algorithms, flexible cluster extraction from dendrograms and various other cluster analysis algorithms. Julia has an implementation inside the Clustering.jl package. Octave, the GNU analog to MATLAB implements hierarchical clustering in function "linkage". Orange, a data mining software suite, includes hierarchical clustering with interactive dendrogram visualisation. R has built-in functions and packages that provide functions for hierarchical clustering. SciPy implements hierarchical clustering in Python, including the efficient SLINK algorithm. scikit-learn also implements hierarchical clustering in Python. Weka includes hierarchical cluster analysis.

5.0 Content Block

The probability that candidate clusters spawn from the same distribution function (V-linkage). The product of in-degree and out-degree on a k-nearest-neighbour graph (graph degree linkage). The increment of

some cluster descriptor (i.e., a quantity defined for measuring the quality of a cluster) after merging two clusters.

6.0 Content Block

In practice, the selection of the number of clusters may also depend on domain knowledge or application-specific requirements. Quantitative measures such as the silhouette coefficient can be used to evaluate the quality of different clusterings. Because hierarchical clustering is sensitive to the choice of distance metric and linkage criterion, the optimal number of clusters may vary depending on these factors. This method builds the hierarchy from the individual elements by progressively merging clusters. In our example, we have six elements $\{a\}$ $\{b\}$ $\{c\}$ $\{d\}$ $\{e\}$ and $\{f\}$. The first step is to determine which elements to merge in a cluster. Usually, we want to take the two closest elements, according to the chosen distance. Optionally, one can also construct a distance matrix at this stage, where the number in the i -th row j -th column is the distance between the i -th and j -th elements. Then, as clustering progresses, rows and columns are merged as the clusters are merged and the distances updated. This is a common way to implement this type of clustering, and has the benefit of caching distances between clusters. A simple agglomerative clustering algorithm is described in the single-linkage clustering page; it can easily be adapted to different types of linkage (see below). Suppose we have merged the two closest elements b and c , we now have the following clusters $\{a\}$, $\{b, c\}$, $\{d\}$, $\{e\}$ and $\{f\}$, and want to merge them further. To do that, we need to take the distance between $\{a\}$ and $\{b, c\}$, and therefore define the distance between two clusters. Usually the distance between two clusters A $\{\displaystyle \mathcal{A}\}$ and B $\{\displaystyle \mathcal{B}\}$ is one of the following:

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Overview HAL-x is a hierarchical density clustering algorithm introduced for large, high-dimensional point clouds. Rather than agglomerating solely from a fixed linkage criterion on a precomputed distance matrix, HAL-x builds an initial set of high-density “pure” clusters in a low-dimensional embedding of a downsampled subset of the data, then merges clusters using supervised linkage: it trains supervised classifiers on the original (unreduced) feature space to measure how reliably two proposed clusters can be separated, and uses those separation scores to decide merge order. The method outputs a hierarchy that can be read as a tree of merges together with a predictive model that can assign labels to held-out or newly collected cells without re-running the entire clustering pipeline on every point during exploratory iterations. In outline, the published implementation proceeds by (i) normalizing and downsampling for dimensionality reduction (commonly t-SNE or UMAP), (ii) estimating density in the embedded space to obtain initial clusters, (iii) constructing a sparse k -nearest neighbor graph between clusters with edge weights derived from classifier-based separability on unreduced coordinates, and (iv) agglomeratively merging the weakest edges while retaining cross-validated accuracy information so users can obtain multiple resolutions from one fitted model.

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Scalable implementation with HAL-x Standard hierarchical agglomerative clustering is difficult to apply to very large datasets because typical implementations require substantial memory for pairwise structures and can scale poorly as n $\{\displaystyle n\}$ grows (see § Complexity above). In computational biology, single-cell technologies such as mass cytometry routinely produce tens of millions of high-dimensional observations, which motivates specialized workflows that preserve biological resolution without requiring that every expensive step be run on the full matrix at once.